

REVALDO FERGUSON

Junior Software Engineer | Full-Stack Developer

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PROFESSIONAL SUMMARY

Junior software engineer building full-stack web applications with Next.js, React, TypeScript, Node.js, Python and Java. Currently contributing Java, SQL and automation work to a live e-commerce platform at Loot.co.za, alongside AWS and Linux infrastructure support. Independently designs, builds and deploys Next.js websites and web apps on Vercel for clubs and small businesses, including an academy admin system and a browser-based barcode scanner. Backed by hands-on DevOps practice with Docker, GitHub Actions CI/CD, Kubernetes, Terraform and Prometheus/Grafana, and 17+ earlier projects covering e-commerce, dashboards, AI assistants and secure authentication. Eduvos IT Software Development graduate (NQF Level 5, February 2026), continuing to a BSc in Information Systems (Software Engineering).

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, SQL, HTML5, CSS3, Bash

Frontend: Next.js, React, Tailwind CSS, responsive and mobile-first UI, accessibility basics

Backend: Node.js, Express, FastAPI, Flask, REST API design, Next.js API routes, Socket.io, authentication and 2FA, Stripe payments

Databases: PostgreSQL, MySQL, MongoDB, data modelling, query optimisation, database administration

DevOps & Cloud: Git and GitHub, GitHub Actions CI/CD, Docker, Kubernetes, Terraform, Nginx, Vercel, AWS (EC2, Elastic Beanstalk), Linux (Red Hat/CentOS)

Testing & Tooling: Python unit testing, automated pipelines, Zabbix, Prometheus and Grafana, Jira, Postman, VS Code

Engineering: OOP (inheritance, polymorphism, method overriding), data parsing and validation, regex, exception handling, debugging, root cause analysis, technical documentation, Agile collaboration

PROFESSIONAL EXPERIENCE

Systems Administrator / Infrastructure | Loot.co.za, Cape Town

September 2025 – Present

Software engineering

- Maintain and extend Java-based e-commerce applications, internal tools and supplier integrations, delivering changes through development tickets.
- Build and maintain supplier data parsing routines, spreadsheet column mappings and regex matching for stock imports and feed changes.
- Apply Java OOP (inheritance, polymorphism, method overriding) with data parsing, validation and exception handling.
- Build automated CSV and reporting tools and automate repetitive business tasks using SQL and database technologies.
- Investigate software and data-processing failures, trace root causes and ship corrective code changes.

Infrastructure and support

- Manage Linux-based on-premise and AWS environments (Red Hat/CentOS, SSH, Systemd); support EC2 and Elastic Beanstalk workloads, including an EC2 resize from t3a.small to t3a.medium.
- Provide 2nd and 3rd line support across infrastructure, applications, operating systems, hardware and connectivity.
- Monitor systems and networks with Zabbix, htop, SSH and Termius; diagnose recurring incidents and root causes.
- Support NIS/ypbind authentication and Postfix email services (mailboxes, quotas, mail queues).
- Manage Microsoft 365, Entra ID and Google Workspace user lifecycle and IT assets across 1,000+ devices.

Administrative Clerk | Bidvest Waltons, Cape Town

August 2024 – March 2025

- Captured and maintained documentation accurately in electronic data capture systems.
- Gave first-line technical assistance to staff and coordinated scheduling and professional correspondence.

CLIENT & NEXT.JS PROJECTS

Hope Rugby Academy *Next.js, API routes, admin*

hope-rugby-academy.vercel.app

- Built and deployed an academy website with an admin area and server-side API routes; added a seed script to populate data.

RCW & RCW Scanner *Next.js, ZXing*

rcw-eta.vercel.app • rcw-scanner.vercel.app

- Built a company website with a clients page, plus a companion web app that scans barcodes in the browser using ZXing.

- Retreat RFC** *Next.js*

retreatrfc-full.vercel.app
- Built and deployed a rugby club website with a fixtures-focused navbar.
- Small-business websites** *Next.js, responsive*

Vercel
- Designed and deployed four sites for a café (Corner Coffee), dental practice (Riverside Dental), florist (Bloom & Co) and auto repair workshop (Atlas Auto Repair).

FEATURED PROJECTS

- IT Pro Solutions — E-commerce Marketplace** *React, Node.js, Stripe, Socket.io, SQL*
- Built a full-stack IT services marketplace with product catalogue, shopping cart and checkout flow.
 - Integrated Stripe payments and order handling, plus real-time customer chat using Socket.io.
 - Designed the relational schema and REST API endpoints for products, carts, orders and users.
- J.A.R.V.I.S. — AI Voice Assistant** *Next.js, FastAPI, Python, Groq, Whisper*
- Developed a voice-driven AI assistant with a Next.js front end and FastAPI backend service layer.
 - Implemented speech-to-text with Whisper, LLM responses via Groq, and persistent conversation memory; added system automation and Spotify control through modular command handlers.
- Glass Admin Dashboard & Authentication System** *React, Node.js, PostgreSQL, 2FA*
- Created an admin dashboard with CRUD user management, role-aware views, dark mode and SQL-backed analytics widgets.
 - Implemented a secure authentication flow with hashed credentials, 2FA and sessions, plus behavioural CAPTCHA and Cloudflare Turnstile variants.
- DevOps Portfolio** *Docker, GitHub Actions, Terraform, Kubernetes, Prometheus, Grafana*
- Cut a Flask container image by 84% with a multi-stage Docker build; ran a 5-replica Nginx deployment on Minikube with self-healing pods.
 - Automated Python tests on every push with GitHub Actions, provisioned Docker containers with Terraform and built a Prometheus and Grafana monitoring stack.

EDUCATION

- BSc Information Systems — Software Engineering** | Eduvos

Planned start: 2027
- IT Software Development, NQF Level 5** | Eduvos, Mowbray Campus

Graduated: February 2026
- National Senior Certificate, NQF Level 4** | Crestway High School

Completed

CERTIFICATIONS & CONTINUOUS LEARNING

- Java Programming, Object-Oriented Programming & Exception Handling (October 2023); Advanced SQL Server diploma (October 2023).
- Diploma in Using Python for Data Science (November 2023); Introduction to Data Analytics with Python (October 2023); Diploma in Microsoft Power BI (November 2023).
- JavaScript development and full-stack coursework (Alison, Code with Mosh).

References available on request.